

Gavin Whitaker

Curriculum Vitae

Contact Information	28 Coronation Avenue Cuffley, Potter's Bar EN6 4FH	Mobile: 07891160747 Email: gavin.whitaker88@gmail.com Website: https://gawhitaker.github.io
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Educational Background	Newcastle University	September 2011 – July 2016
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Qualification	PhD in Statistics
Thesis Title	Bayesian Inference for Stochastic Differential Mixed-effects Models
Supervisors	Dr Andrew Golightly and Prof Richard Boys

Newcastle University	September 2007 – June 2011
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Qualification	Master of Mathematics and Statistics (MMathStat)
Grade	First Class Honours
Masters Title	The Bivariate Poisson Distribution and its Applications to Football
Supervisor	Dr Philip Ansell

Scarborough Sixth Form College	September 2005 – June 2007
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4 A Levels: Mathematics (A), Further Mathematics (B), History (B), Drama and Theatre Studies (B).

George Pindar Community Sports College	September 2000 – June 2005
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10 GCSE's at A*-C including A's in Mathematics, Science and English. Full list available on request.

Technical Skills	Statistical Skills
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I am interested in, and have experience working with: Bayesian statistics, sports modelling, Markov chain Monte Carlo methods, mixed/random effects modelling, Bayesian hierarchical models, variational inference, time series analysis, and stochastic differential equations.

Computer Packages

R - I am an experienced user of the Statistical package R and write my own functions/packages.

Python - I have experience with Python, writing my own functions and classes, along with utilising packages such as `pystan` and `pymc3`.

C - I have experience with the C programming environment, gained from requiring greater efficiency in the codes I used for my PhD. This includes parallelisation through the use of `openmp`.

I am experienced with Microsoft Office and proficient with all its elements, word processing, spreadsheets, presentations etc.

I also have experience with numerous other computer packages such as LaTeX (Typesetting program for preparing documents and presentations), Minitab (Statistical software), HTML (HyperText Markup Language) and SVN and Git (Version control).

Operating Systems

I am very familiar with Microsoft Windows and Ubuntu Linux operating systems. I have experience in programming from the command line in Ubuntu.

Roles and Responsibilities

I was a committee member for the Royal Statistical Society Statistics in Sport Section (2018) and belonged to The Alan Turing Institute "Data science for sports, activity, and well-being" interest group. For both, I helped to organise talks and events, arranging speakers and sessions, along with general organisation.

Problems class leader for undergraduate studies at Newcastle University. I have presented the full set of problems classes (33 hours) for a first year course on "Quantitative Methods for Business Management", consisting of an 11 hour course repeated 3 times for approximately 100 undergraduate students (2013).

Practical demonstrator for undergraduate studies at Newcastle University. I have demonstrated for several courses using statistical packages R and Minitab.

(Selected) Publications

Whitaker GA, Silva R, Edwards D. "A Bayesian inference approach for determining player abilities in soccer." In submission.

Whitaker GA, Silva R, Edwards D. "Visualizing a team's goal chances in soccer from attacking events: a Bayesian inference approach." *Big data*, 2018, 6(4), 271-290.

Tsokos A, Narayanan S, Kosmidis I, Baio G, Cucuringu M, Whitaker GA, Király FJ. "Modeling outcomes of soccer matches." *Machine Learning*, 2018, 1, 77-95.

Whitaker GA, Golightly A, Boys RJ, Sherlock C. "Improved bridge constructs for stochastic differential equations." *Statistics and Computing*, 2017, 27(4), 885-900.

Whitaker GA, Golightly A, Boys RJ, Sherlock C. "Bayesian inference for diffusion-driven mixed-effects models." *Bayesian Analysis*, 2017, 12(2), 435-463.

(Selected) Presentations

July 2018, ISBIS, University of Piraeus, Greece:
Using player abilities to predict football.

June 2018, ISBA 2018, Edinburgh (poster presentation):

A Bayesian inference approach for determining player abilities in football.

December 2017, RSS local group seminar, Newcastle University:
Using player abilities to predict football.

June 2016, ISBA 2016, Sardinia (poster presentation):
Bayesian inference for diffusion driven mixed-effects models.

April 2014, Research Students Conference, Nottingham University:
Bayesian inference for stochastic differential random effects models (winning a prize for best talk).

Recent Employment

Quantitative Analyst
Smartodds

May 2021 – June 2026

I worked as a quantitative analyst for Smartodds in my last role. I modelled sports, primarily cricket and golf, with the aim of predicting the outcomes. To do this I helped with building in-house models which took data from various sources and using such techniques as Bayesian hierarchical modelling, GAM, and state-space modelling created predictions. This may have been for match outcomes, or for the first innings totals in cricket, for example. Presenting ideas internally at informal seminars was part of the role to spread knowledge. Code creation (fuctions/packages in R), maintance and documentation was expected, along with problem solving and the ability to think quickly for live issues. Outside of this role I also oversaw the supervision of masters projects that the company worked on with various universities.

Data Scientist
BDO LLP

November 2018 – April 2021

I worked as a data scientist for BDO, starting, and helping to set up, their data science offering whilst working on a variety of projects. I have worked with the energy sector evaluating the fairness of schemes, and providing the data analytics for value for money assessments. I have built outlier detection mechanisms into auditing software to aid the detection of fraud, helped motor insurers to better understand their pricing models and how policies are priced, and refined models to determine earthquakes and their damage. Working with property management companies I have created rating systems for their surveyors. Data analytics have also been performed for international charity projects to aid systems that are often large scale and difficult to monitor. Results have been presented in both a technical and non-technical manner to ensure all have access to findings and where they come from.

References

Available on request.